

7.7 Polymerisation (A Level Only)

Question Paper

Course	CIE A Level Chemistry
Section	7. Organic Chemistry (A Level Only)
Topic	7.7 Polymerisation (A Level Only)
Difficulty	Hard

Time allowed: 30

Score: /24

Percentage: /100

Question 1a

Polymers consist of monomers joined together by undergoing either addition or condensation polymerisation.

Fig. 1.1 shows a dicarboxylic acid and a diol that can be used in a general condensation polymerisation reaction.

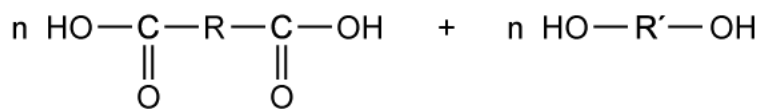


Fig. 1.1

Draw **one** repeat unit for this polymerisation.

repeat unit

[1 mark]

Question 1b

In terms of n , state the number of molecules of water formed in the condensation polymerisation reaction of a general dicarboxylic acid and a general diol as shown in Fig. 1.1.

[1 mark]

Question 1c

Using displayed formulae, write the balanced equation for the condensation polymerisation of the two monomers, propanedioyl dichloride and butane-1,4-diol.

[3 marks]

Question 1d

Suggest an alternative pair of monomers that will produce the polymer from part (c).

[1 mark]

Question 2a

Lactic acid has the structural formula, $\text{CH}_3\text{CHOHCOOH}$.

i)

Name all the functional groups in lactic acid.

[1]

ii)

Give the systematic name of lactic acid.

[1]

iii)

Lactic acid exhibits stereoisomers.

Draw three-dimensional structures for the two stereoisomers of lactic acid.

Name this type of stereoisomerism.



type of stereoisomerism

[2]

[4 marks]

Question 2b

Poly(lactic acid) is a thermoplastic polyester that can be prepared from lactic acid. It is a renewable material used widely in 3D printers.

Explain why lactic acid can be polymerised.

[1 mark]

Question 2c

Poly(lactic acid) contains methyl groups.

i)

Draw the **skeletal** formula for **three** repeat units of poly(lactic acid).

[1]

ii)

Explain what happens to the chirality of lactic acid when it is polymerised.

[1]

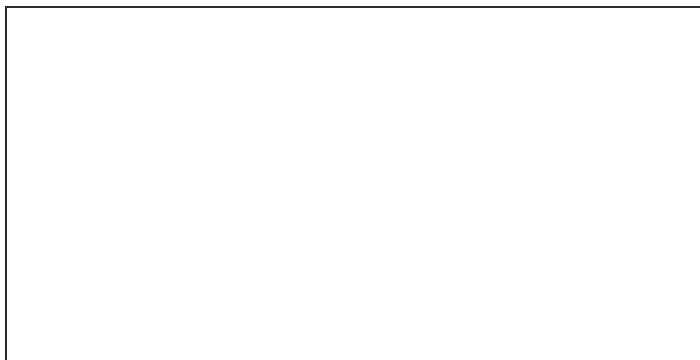
[2 marks]

Question 2d

Poly(lactic) acid can be prepared using a different monomer.

Draw the **skeletal** formula of this monomer.

Give the systematic name of another monomer that could make poly(lactic acid).



systematic name

[2 marks]

Question 3a

One family of condensation polymers are the aramids, so called because of the arene rings that are linked together with amide bonds. Kevlar[®] and Nomex[®] are two common examples of aramids.

Kevlar[®] has a variety of uses, including bulletproof vests and puncture resistant tyres, due to its strength and being lightweight. Nomex[®] is most commonly used as a lining of the overalls worn by racing drivers and their pit crews as well as in the personal protective equipment of firefighters due to its flame resistant properties.

Nomex[®] is made from 1,3-diaminobenzene and 1,3-benzenedicarboxylic acid.

Draw the **skeletal** structures of these monomers.



[2 marks]

Question 3b

Draw the structure of the Nomex[®] polymer.

[3 marks]

Question 3c

Kevlar[®] is made from 1,4-diaminobenzene and 1,4-benzenedicarboxylic acid.

Draw **two** repeat units of the polymer to show the strongest intermolecular force in Kevlar[®].

[2 marks]

Question 3d

State whether Kevlar[®] or Nomex[®] will have the higher melting point.

Explain your answer.

[2 marks]

Model Answers: Hard

1a

a) **One** repeat unit for the general condensation polymerisation of a dicarboxylic acid and a diol is:

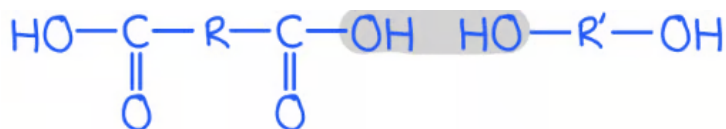
- Correct structure of the repeat unit

AND

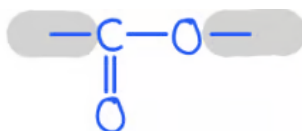
Continuation bonds; [1 mark]

[Total: 1 mark]

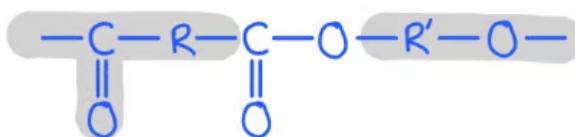
- **Remember:** The condensation polymerisation of a dicarboxylic acid and a diol is basically a series of esterifications



- The repeat unit **must** come from the esterification of a carboxylic acid group and an alcohol group



- Then you attach the corresponding R-groups and complete your repeat unit



- **Remember:** The repeat unit does not have the carboxylic acid and alcohol complete, they should be incomplete so that they can react

1b

b) In terms of n , the number of molecules of water formed in the condensation polymerisation reaction of a general dicarboxylic acid and a general diol is:

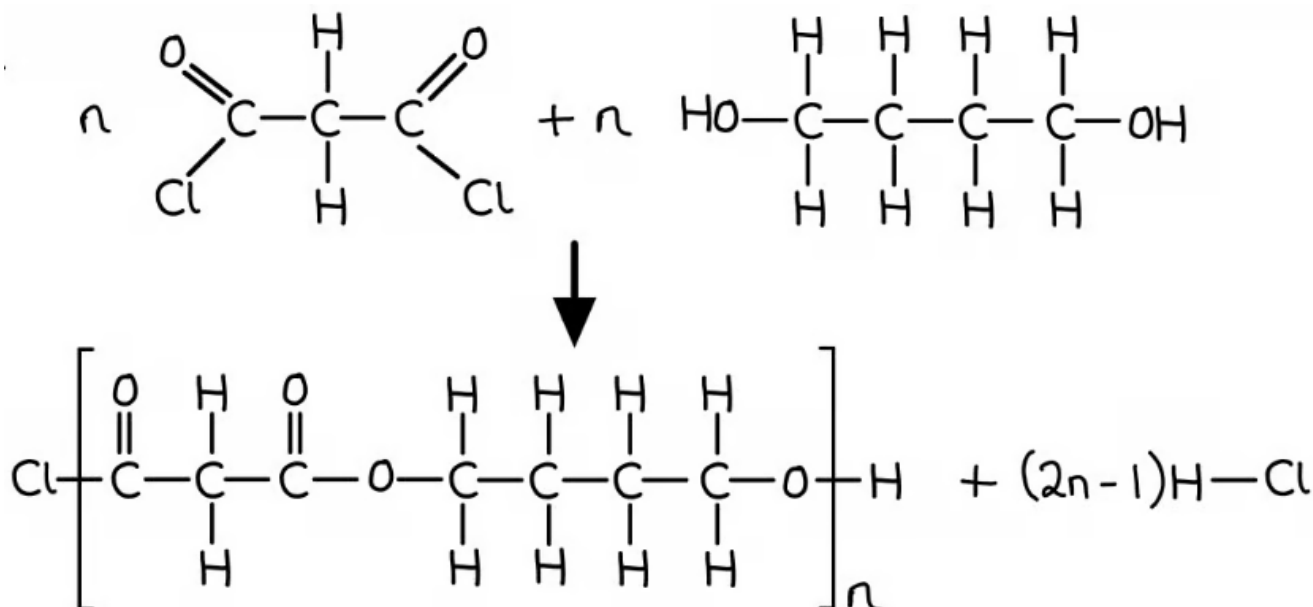
- $2n - 1$; [1 mark]

[Total: 1 mark]

- Using the phrase “*in terms of n* ” makes this question sound more challenging than it is and may cause you to give up on this question unnecessarily
 - Start simple, if $n = 1$
 - This means that one acid is reacting with one alcohol
 - Therefore, one ester is formed with the release of one water
 - Now, try $n = 2$
 - This means that the polymer formed is acid - alcohol - acid - alcohol
 - There are three ester links meaning that three waters are released
 - If you need to, try $n = 3$
 - This means that the polymer formed is acid - alcohol - acid - alcohol - acid - alcohol
 - There are five ester links meaning that five waters are released
 - You now have the following information to deduce the number of molecules of water
 - $n = 1$ gives one water molecule
 - $n = 2$ gives three water molecules
 - $n = 3$ gives five water molecules
 - You have to double n , then take away one
 - Therefore, the number of molecules of water formed is $2n - 1$

1c

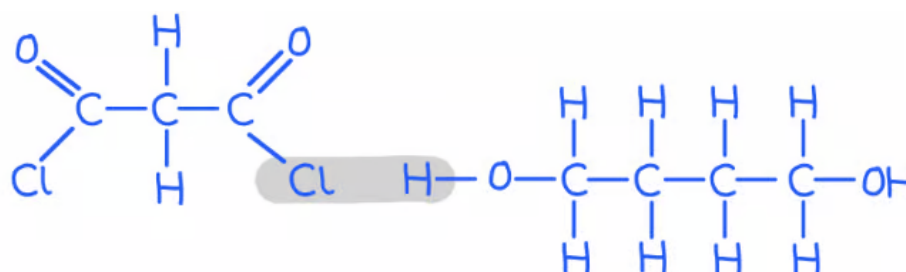
c) Using displayed formulae, the balanced equation for the condensation polymerisation of the two monomers, propanedioyl dichloride and butane-1,4-diol is:



- Correct displayed formula of propanedioyl dichloride, including n
AND
Correct displayed formula of butane-1,4-diol, including n ; [1 mark]
- Correct structure of the polyester formed including brackets and n ; [1 mark]
- $(2n - 1)$ HCl; [1 mark]

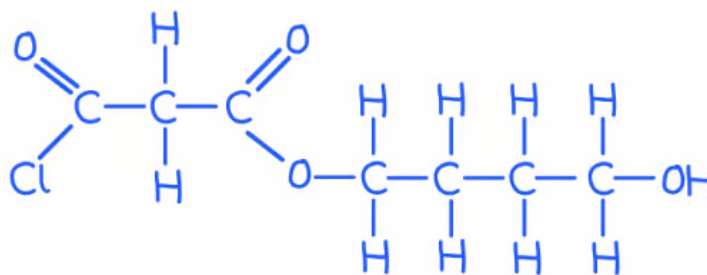
[Total: 3 marks]

- You are expected to be able to draw butane-1,4-diol as well as deduce the structure of propanedioyl dichloride
- Propanedioyl dichloride suggests a propane chain with 2 (di) acyl chloride functional groups
- You are told that this is a condensation polymerisation reaction
 - Draw the two monomers with the reacting functional groups next to / near to each other



- The question specifies displayed formulae, so you will lose the mark if you do not draw the structures as a displayed formula in your answer

- Remove the condensation product and replace this section with a bond to complete the ester link



- **Remember:** For a general condensation polymerisation reaction, there is always $2n - 1$ moles of the condensation product (as shown in part (b))

1d

d) An alternative pair of monomers that will produce the polymer from part (c) are:

- Propanedioic acid

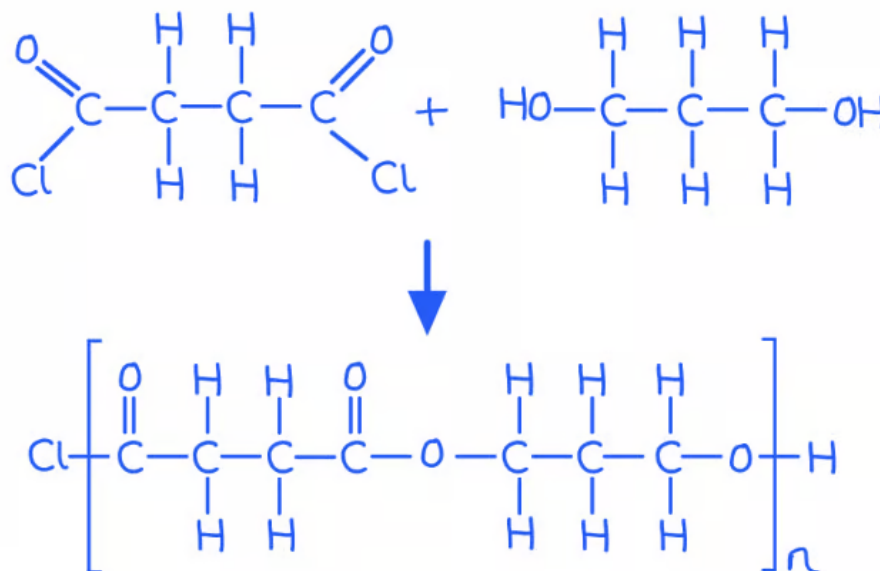
AND

Butane-1,4-diol; [1 mark]

[Total: 1 mark]

- The condensation polymerisation featured in part (c) is a series of esterification steps
- Part (c) uses a diacyl chloride with butane-1,4-diol
- Therefore, one possible suggestion would be the use of the dicarboxylic acid with butane-1,4-diol
 - There might be other possibilities for this reaction but the chemicals involved are likely to be beyond the scope of A-level Chemistry
- **Careful:** You might simply try and swap the diacyl chloride and diol over, e.g. butanedioyl dichloride and propane-1,3-diol but this does not produce the same

- There might be other possibilities for this reaction but the chemicals involved are likely to be beyond the scope of A-level Chemistry
- **Careful:** You might simply try and swap the diacyl chloride and diol over, e.g. butanedioyl dichloride and propane-1,3-diol but this does not produce the same condensation polymer



2a

a)

i) The functional groups in lactic acid are:

- Alcohol / hydroxyl

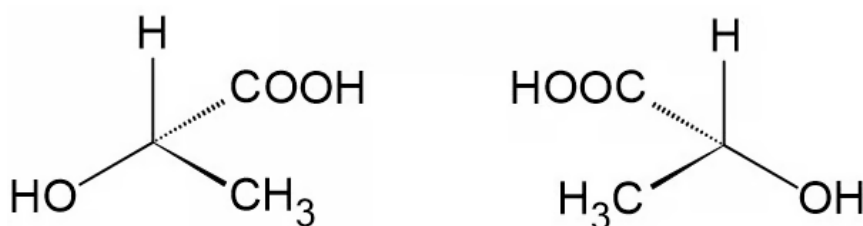
AND

Carboxylic acid / carboxyl; [1 mark]

ii) The systematic name of lactic acid is:

- 2-hydroxypropanoic acid; [1 mark]

iii) The three-dimensional structures for the two stereoisomers of lactic acid are:



- Two structures with the correct functional groups
AND
The structures are mirror images of each other; [1 mark]

The type of stereoisomers is:

- Optical (isomerism); [1 mark]

[Total: 3 marks]

- For part (i)
 - The functional groups are:
 - OH attached to the second carbon - alcohol / hydroxyl
 - The COOH at the end of the structural formula - carboxylic acid / carboxyl
- For part (ii)
 - From the structural formula you can see:
 - There are three carbon atoms in total → prop
 - There is a carboxylic acid group → oic acid, with this carbon being carbon-1
 - There is a hydroxyl group on carbon-2 → 2-hydroxy
 - Putting all of these pieces gives the systematic name 2-hydroxypropanoic acid
- For part (iii)
 - The second carbon is the chiral centre as it has four different atoms / groups of atoms attached
 - Methyl / CH₃
 - Hydrogen / H
 - Alcohol / Hydroxyl / OH
 - Carboxylic acid / carboxyl / COOH
 - This means that you are dealing with optical isomerism
 - To draw the three-dimensional structures:
 1. Place a central chiral carbon atom
 2. Draw two straight-line / "in the plane" bonds, one full wedge and one dotted wedge bond off the central chiral carbon atom
 3. Place one different atom / group of atoms at the end of each bond
 - **Careful:** Make sure that the correct atom is touching the bond e.g. for the alcohol / hydroxyl / OH group, the bond must be clearly going to the O

4. Draw the mirror image of your first three-dimensional structure

2b

b) Lactic acid can be polymerised because:

- It contains an alcohol / hydroxyl / OH group

AND

A carboxylic acid / carboxyl / COOH group

AND

That can undergo condensation polymerisation; [1 mark]

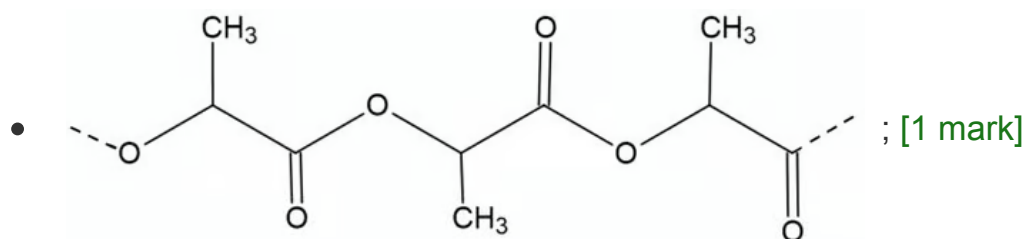
[Total: 1 mark]

- The question gives you a clue when it says that poly(lactic) acid is a polyester
- Remember:** Polyesters are formed by condensation reactions involving alcohol / hydroxyl / OH groups and carboxylic acid / carboxyl / COOH groups

2c

c)

i) A displayed formula showing three repeat units of poly(lactic acid) is:



ii) When lactic acid is polymerised, the chirality:

- Remains / stays

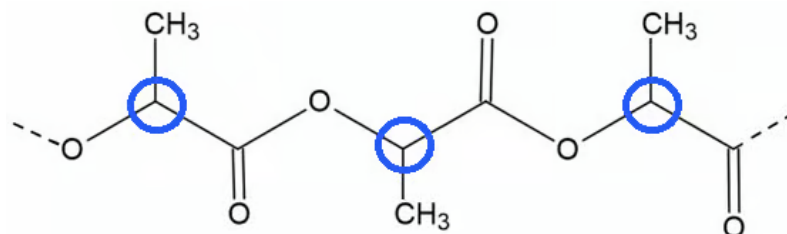
AND

(Because,) the polymer / poly(lactic) acid still has a chiral centre; [1 mark]

[Total: 2 marks]

- For part (i)
 - You must show the continuation bonds for the mark

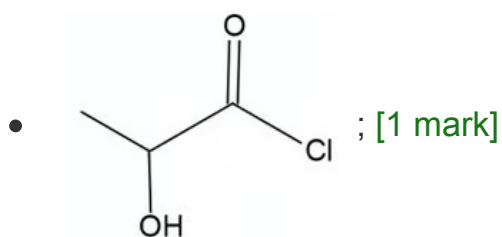
- You can show a straight chain with all the side groups on the same side, but the most likely configuration is shown in the diagram with the side groups alternating to give the largest repulsion angle
- For part (ii)
 - The chirality remains because every other carbon is attached to four different atoms / groups of atoms:



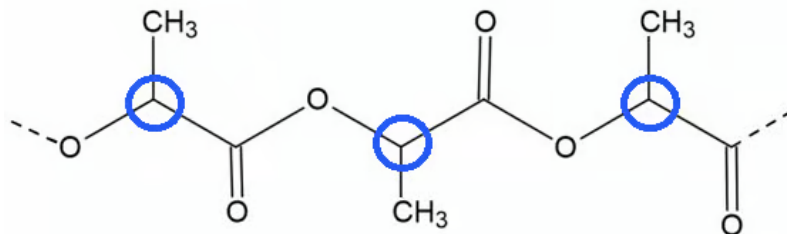
- The four different atoms / groups of atoms are:
 - Single bond to oxygen
 - Methyl / CH₃ group
 - Single bond to C=O
 - "Hidden" bond to hydrogen
 - **Remember:** Carbon typically does four bonds but there should only be three showing in your answer to part (i)

2d

d) Another monomer that could make poly(lactic) acid is:



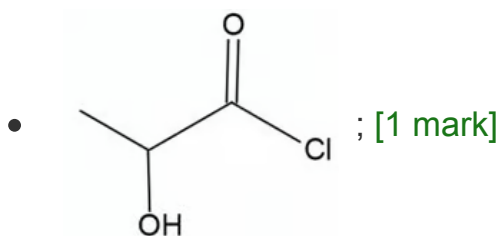
- The chirality remains because every other carbon is attached to four different atoms / groups of atoms:



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 - Single bond to oxygen
 - Methyl / CH₃ group
 - Single bond to C=O
 - "Hidden" bond to hydrogen
 - **Remember:** Carbon typically does four bonds but there should only be three showing in your answer to part (i)

2d

d) Another monomer that could make poly(lactic) acid is:



- 2-hydroxypropanoyl chloride; [1 mark]

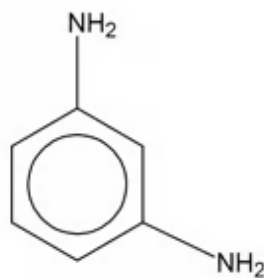
[Total: 2 marks]

- **Remember:** Generally, acid chlorides / acyl chlorides will undergo the same reactions as carboxylic acids, but more vigorously!
- Therefore, if you can make polyesters from hydroxycarboxylic acids, then you can also make polyesters from hydroxy acid chlorides

3a

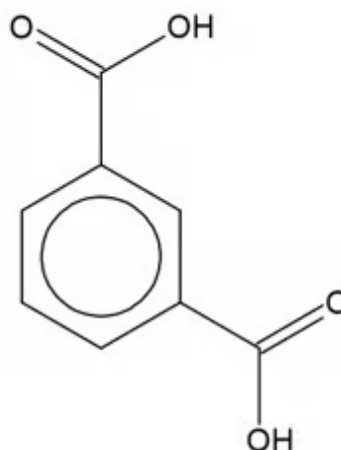
a) The skeletal structures of 1,3-diaminobenzene and are:

- 1,3-diaminobenzene =



; [1 mark]

- 1,3-benzenedicarboxylic acid =



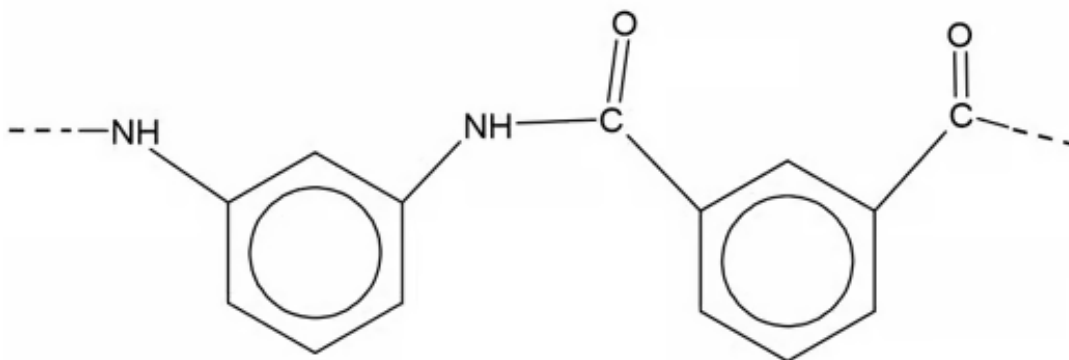
; [1 mark]

[Total: 2 marks]

- Both monomers are based on the benzene ring and have functional groups attached at the 1 and 3 position
- **Careful:** A common mistake is to draw the functional groups in the 1 and 4 positions, this possibly happens due to the habit of spreading functional groups as far apart as possible
- The amine group is -NH_2
- The carboxylic acid group is -COOH
- Both monomers are di-substituted which means that they have 2 of each functional group attached

3b

b) The structure of the Nomex[®] polymer is:



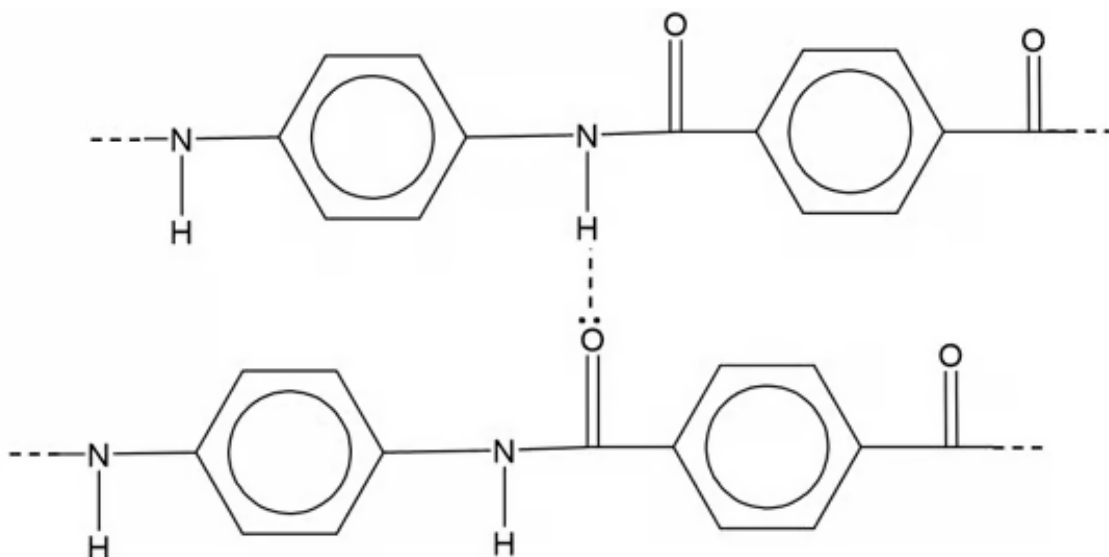
- Correct amide link of the monomers; [1 mark]
- Correct repeat unit of the polymer; [1 mark]
- The polymer structure includes an amino / NH group at one end
AND
A carboxyl group at the other end; [1 mark]

[Total: 3 marks]

- **Remember:** This is the condensation polymerisation of a diamine and a dicarboxylic acid, so the products will be the polyamide and water
- Place the monomers next to each other
- Identify the atoms that can make the water molecule

3c

c) Two repeat units of the polymer to show the strongest intermolecular force in Kevlar®:



There are other variations with the monomers the other way around / a mixture of the other monomers

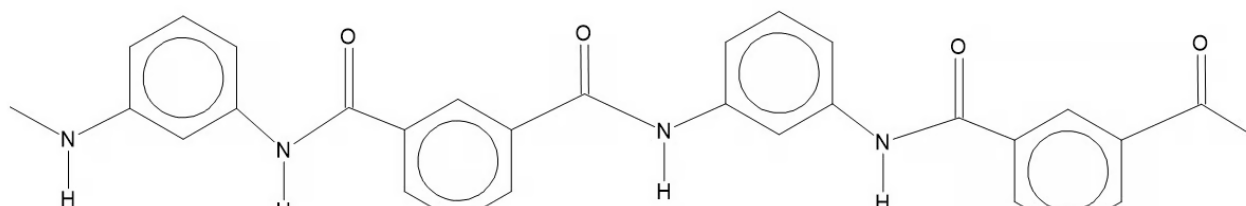
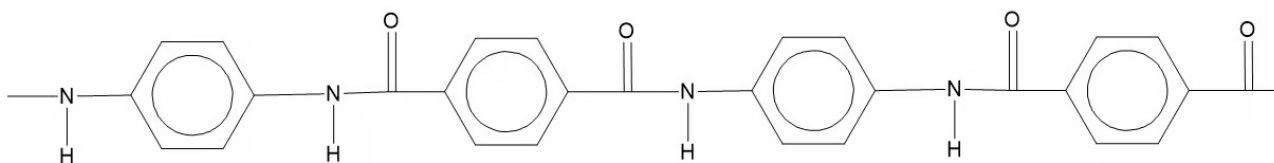
- Correct structure of both repeat units; [1 mark]
- Correct hydrogen bonding between the amide links of the repeat units, starting at the lone pair of the oxygen; [1 mark]

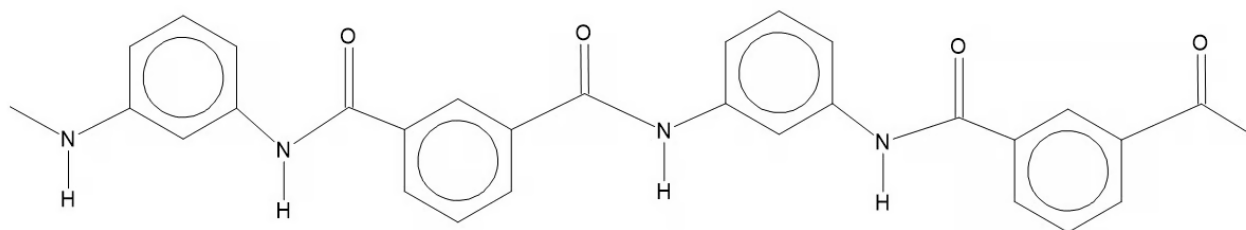
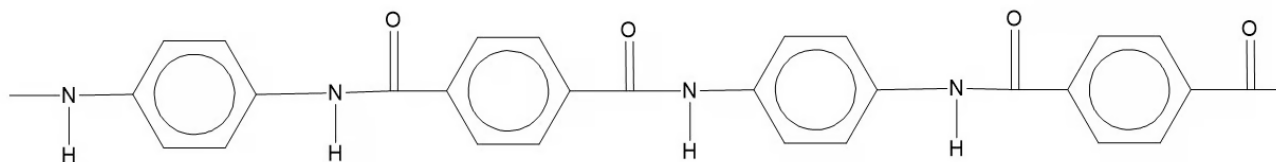
[Total: 2 marks]

- As the question states “strongest intermolecular force” you should automatically be thinking about hydrogen bonding
- The hydrogen bonding must occur between the amide links of the polymer:
 - The lone pair of the oxygen
 - The hydrogen attached to the nitrogen
- Whenever you draw a hydrogen bond, regardless of the topic, it should always start at the lone pair

3d

d) The polyamide with the higher melting point will be:





- Kevlar; [1 mark]

Explanation:

- The polymer chain of Kevlar is straighter
AND
Therefore the molecules / polymer chains lie closer together
AND
The intermolecular forces are / hydrogen bonding is more effective; [1 mark]

[Total: 2 marks]

- You can give the reverse argument for Nomex[®] being more zig-zag shaped, molecules lying further apart and having less effective intermolecular forces
- This question relies on you recognising the key difference between the polymer and monomer structures of Kevlar[®] and Nomex[®]
Kevlar[®] is all 1,4-substitution on the benzene ring - it is straight
- Nomex[®] is all 1,3-substitution on the benzene ring - it is not straight
- Everything else about the composition of the monomers and polymers is the